1.5 Measuring and Constructing Angles

Name _____ Period ____ Date _____

Geometry 1.5 (30-minute plan) · Student sheet

I can…

- name, measure, and classify angles.
- use the Angle Addition Postulate to find angle measures.
- use an angle bisector to find angle measures.

Vocabulary · fill in the missing word

Angle — a figure formed by two rays with the same _____, called the vertex

Acute angle — an angle that _____ between 0° and 90°

Right angle — an angle that _____ exactly 90°

Obtuse angle — an angle that _____ between 90° and 180°

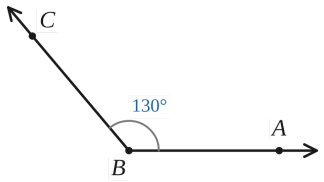
Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Notes and examples (fill in each note, then work the example)

Naming angles: ∠ABC, ∠CBA, and ∠B can name the same angle; the vertex _____ is the middle letter.

Example 1

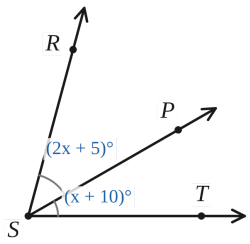
Name the angle in three ways. Then classify it.



Angle Addition Postulate: if P is in the interior of ∠RST, then m∠RSP + m∠PST = _____.

Example 2

m∠RST = 75°. Find x, m∠RSP, and m∠PST.



Angle bisector: it divides an angle into two _____ angles, so each half is half the whole.

Example 3

Ray YW bisects ∠XYZ. m∠XYW = (4x − 3)° and m∠WYZ = (2x + 11)°. Find x and m∠XYZ.

Part 2 · Partner practice

Partner practice

A ray bisects an angle into ∠1 and ∠2. m∠1 = (5x − 4)° and m∠2 = (3x + 12)°. Find x and the measure of the whole angle.